



30TH INTERNATIONAL SYMPOSIUM ON VLSI DESIGN AND TEST (VDAT- 2026)

Advancing Intelligent, Secure and Quantum-Ready Semiconductor Systems
for the Next Era of Computing

20 - 22 August 2026

Jaypee Institute of Information Technology, Sector-62, Noida

CALL FOR PAPERS

All accepted and presented papers will be submitted for inclusion in the IEEE Xplore Digital Library, subject to IEEE quality review and compliance.

VDAT 2026 invites original, high-quality research papers in the areas of VLSI design, semiconductor devices, AI-driven hardware systems, advanced packaging, reliability, security, and quantum computing architectures. The symposium aims to bring together academia, industry professionals, researchers, and innovators to discuss emerging trends and technological advancements shaping the future of intelligent semiconductor systems. Authors are encouraged to submit unpublished work presenting novel methodologies, design techniques, experimental results, and system-level innovations.



VLSI & System-on-Chip (SoC) Design

- Digital, Analog & Mixed-Signal Design
- Low-Power & High-Performance Architectures
- Heterogeneous SoC Integration
- Embedded & Real-Time Systems
- Memory Subsystems & On-Chip Interconnects
- Embedded SoC Architectures for Robotics and Autonomous Systems



Nanoelectronics, Emerging Devices & Beyond-CMOS Technologies

- FinFET, GAA, Nanosheet & Advanced Transistor Structures
- 2D Materials & Novel Semiconductor Devices
- Spintronics & Neuromorphic Devices
- Device Modeling & Reliability
- Cryogenic & Advanced Device Technologies



Electronic Design Automation (EDA), CAD & AI-Driven Design

- Physical Design & Place-and-Route
- Design Space Exploration
- AI/ML for EDA & Automation
- Statistical & Formal Verification
- Design Automation for Advanced Technology Nodes
- AI-Driven Hardware Design Optimization for Robotic Control Systems



AI Hardware & VLSI for AI/ML Workloads

- Hardware Accelerators for Deep Learning
- Edge AI & Low-Power AI Systems
- In-Memory & Near-Memory Computing for AI
- AI for Semiconductor Manufacturing & Yield
- Efficient Architectures for LLMs & Emerging AI Models
- Edge AI Hardware Accelerators for Robotics and Autonomous Platforms



Verification, Testing, Reliability & Hardware Security

- Design-for-Testability (DFT) & Built-In Self-Test (BIST)
- Hardware Trust & Security Architectures
- Fault Tolerance & Resilient Systems
- Aging, Variability & Radiation Effects
- Secure SoC & Cryptographic Hardware



3D-IC, Advanced Packaging & Heterogeneous Integration

- 2.5D/3D IC Integration
- Chiplets & System-in-Package
- Thermal & Power Integrity in 3D Systems
- Advanced Interconnect & TSV Technologies
- Packaging Solutions for AI & High-Performance System



Quantum and Future Computing Architectures

- Quantum Computing Devices & Architectures
- Quantum Circuit Design & Error Correction
- Cryogenic CMOS for Quantum Systems
- Beyond-CMOS Computing Paradigms
- Hybrid Classical-Quantum Architectures

SCAN TO SUBMIT



<https://softconf.com/vlsi/vdat2026>

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SUBMISSION INFO

Authors are invited to submit full-length (6 pages maximum), original, unpublished papers along with an abstract of at most 200 words. Paper format: Submissions should be in camera-ready IEEE two-column format, following the IEEE proceedings specifications.

IMPORTANT DATES

30th April
2026

Paper
submission
deadline

30th June
2026

Acceptance
notification

6th July
2026

Early bird
registration
and payment

16th July
2026

Regular
registration
and payment

20-22nd August
2026

Conference
date

